

VISION

To achieve academic excellence in Information Technology by providing quality education, and training in the latest technologies to yield innovative, technically competent, and socially responsible professionals to meet global challenges.

MISSION

To impart knowledge in a way that caters to the needs of the ever-changing technologies.

To exhibit trained competent professionals in the design and development of software products to fulfil industry needs.

To infuse a sense of ethics and social responsibilities among the students.

PROGRAM EDUCATIONAL OBJECTIVES (PEO)

1. To educate students efficiently in core areas of Information Technology.
2. The graduates of the program will acquire technical and research in the fields of Information Technology.
3. Aim is to have Graduates who achieve lifelong learning through successful completion of higher education.
4. Produce graduates who will perform their duty ethically and responsibly in their profession.

PROGRAM SPECIFIC OBJECTIVES (PSO)

1. To understand, analyze and develop computer programs in the areas related to algorithms, system software, multimedia, web design, DBMS, and networking for efficient design of computer-based systems.
2. To provide solutions for real-world problems with a wide range of programming languages and open-source platforms in various computing domains.
3. To develop an ability to be an entrepreneur.

COURSE OUTCOMES (CO)

1. Write, Test and Debug Python Programs.
2. Use functions and represent Compound data using Lists, Tuples and Dictionaries.
3. Text filtering with regular expressions in Python.
4. Data manipulation and data visualization in Python.
5. Create GUI applications in Python.

Name:		Enrolment No:		Semester:	
Sr. No.	Definition	Date	Page No.	Signature	Remarks
01	Basic Program 1. Study of class path and java runtime environment 2. Write a program to - Implement command line calculator - Write To prints Fibonacci series.				
02	Array: 1. Define a class Array with following member Field: int data[]; Function: Array() //create array data of size 10 Array(int size) // create array of size size Array(int data[]) // initialize array with parameter array void Reverse _an _array () //reverse element of an array int Maximum _of _array () // find maximum element of array int Average_of _array() //find average of element of array void Sorting () //sort element of array void display() //display element of array int search(int no) //search element and return index else return -1 int size(); //return size of an array Use all the function in main method. Create different objects with different constructors. 2. Define a class Matrix with following Field: int row, column; float mat[][] Function: Matrix(int a[][]) Matrix() Matrix(int rwo, int col) void readMatrix() //read element of array float [][] transpose() //find transpose of first matrix float [][] matrixMultiplication(Matrix second) //multiply two matrices and return result				

	<pre>void displayMatrix(float [][]a) //display content of argument array void displayMatrix() //display content float maximum_of_array() // return maximum element of first array float average_of_array() // return average of first array create three object of Matrix class with different constructors in main and test all the functions in main</pre> 3. Write a program to demonstrate usage of different methods of Wrapper class 4. Write a program to demonstrate usage of String and StringBuffer class										
03	Basic Program using Class 1. Create a class BankAccount that has Depositor name , Acc_no, Acc_type, Balance as Data Members and void createAcc() . void Deposit (), void withdraw() and void BalanceInquiry as Member Function. When a new Account is created assign next serial no as account number. Account number starts from 1 2. Create a class time that has hour, minute and second as data members. Create a parameterized constructor to initialize Time Objects. Create a member Function Time Sum (Time, Time) to sum two-time objects. 3. Define a class with the Name, Basic salary, and dearness allowance as data members. Calculate and print the Name, Basic salary(yearly), dearness allowance and tax deduced at source (TDS) and net salary, where TDS is charged on gross salary which is basic salary + dearness allowance and TDS rate is as per following table. <table><tr><td>Gross Salary</td><td>TDS</td></tr><tr><td>Rs. 100000 and below</td><td>NIL</td></tr><tr><td>Above Rs. 100000</td><td>10% on excess over 100000</td></tr></table> DA is 74% of Basic Salary for all. Use appropriate member function.	Gross Salary	TDS	Rs. 100000 and below	NIL	Above Rs. 100000	10% on excess over 100000				
Gross Salary	TDS										
Rs. 100000 and below	NIL										
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04	Inheritance and interface 1. class Cricket having data members name, age and member methods display() and setdata(). class Match inherits Cricket and has data members no_of_odi, no_of_test. Create an array of 5 objects of class Match. Provide all the required data through command line and display the information. 2. Declare an interface called Property containing a method computePrice to compute and return the price. The interface is to be implemented by following two classes i) Bungalow and ii) Flat. Both the classes have following data members - name										

	-constructionArea. The class Bungalow has an additional data member called landArea. Define computePrice for both classes for computing total price. Use following rules for computing total price by summing up sub-costs: Construction cost(for both classes): Rs.500/ - per sq.feet Additional cost (for Flat) : Rs. 200000/- (for Bungalow) : Rs. 200/- per sq. feet for landArea Land cost (only for Bungalow) : Rs. 400/- per sq. feet Define method main to show usage of method computePrice. 3. Define following classes and interfaces. public interface GeometricShape { public void describe(); } public interface TwoDShape extends GeometricShape { public double area(); } public interface ThreeDShape extends GeometricShape { public double volume(); } public class Cone implements ThreeDShape { private double radius; private double height; public Cone (double radius, double height) public double volume() public void describe() } public class Rectangle implements TwoDShape { private double width, height; public Rectangle (double width, double height) public double area() public double perimeter() public void describe() } public class Sphere implements ThreeDShape { private double radius; public Sphere (double radius) public double volume() public void describe() } Define test class to call various methods of Geometric Shape				
05	Inner Class & Packages:				

	Define two nested classes: Processor and RAM inside the outer class: CPU with following data members <pre> class CPU { double price; class Processor{ // nested class double cores; double catch() String manufacturer; double getCache() void displayProcesorDetail() } protected class RAM{ // nested protected class // members of protected nested class double memory; String manufacturer; Double clockSpeed; double getClockSpeed() void displayRAMDetail() } } </pre> 1) Write appropriate Constructor and create instance of Outer and inner class and call the methods in main function Packages: 1) Implement a program that defines package figures, which has a class circle inherited from a class shape in it. Include the methods in the class to calculate area and display the necessary information. Also include a class, which has main to carry out above-mentioned operations. 2) Create a package new_figures, which has classes rectangle and triangle inherited from class shape of package figures. Thus import the package figure. Carry out necessary operations to calculate area and display the necessary information. 3) Modify the above programs by importing both the packages in a file outside both the packages, which contains main class.				
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06	Generics 1. Declare a class InvoiceDetail which accepts a type parameter which is of type Number with following data members class InvoiceDetail <N extends Number> { private String invoiceName; private N amount; private N Discount // write getters, setters, and constructors } Call the methods in Main class 2. Write a program to sort the object of Book class using comparable and comparator interface. (Book class consist of book id, title, author and publisher as data members)				
07	Exception Handling 1. Write a program for creating a Bank class, which is used to manage the bank account of customers. Class has two methods, Deposit () and withdraw (). Deposit method display old balance and new balance after depositing the specified amount. Withdrew method display old balance and new balance after withdrawing. If balance is not enough to withdraw the money, it throws ArithmeticException and if balance is less than 500rs after withdrawing then it throw custom exception, NotEnoughMoneyException. 2. Write a complete program for calculation average of n +ve integer numbers of Array A. a. Read the array form keyboard b. Raise and handle Exception if i. Element value is -ve or non-integer. ii. If n is zero.				
08	Threading 1. Write a program to find prime number in given range using method of multithreading. Also run the same program using executor framework 2. Assume one class Queue that defines queue of fix size says 15. • Assume one class producer which implements Runnable, having priority NORM_PRIORITY +1 • One more class consumer implements Runnable, having priority NORM_PRIORITY-1 • Class TestThread is having main method with maximum priority, which creates 1 thread for producer and 2 threads for consumer. • Producer produces number of elements and put on the queue. when queue becomes full it notifies other threads. Consumer consumes number of elements and notifies other thread when queue become empty.				
09	Collection API:				

	1. Write a program to demonstrate user of ArrayList, LinkedList, LinkedHashMap, and HashSet Class. 2. Write a program to Sort Array, ArrayList, String, List, Map and Set				
10	File Handling Using Java: 1. Write a programme to count occurrence of a given words in a file. 2. Write a program to print it self. 3. Write a program to display list of all the files of given directory				
11	Networking 1. Implement Echo client/server program using TCP 2. Write a program using UDP which give name of the audio file to server and server reply with content of audio file				
12	GUI 1. Write a programme to implement an investement value calculator using the data inputed by user. textFields to be included are amount, year, interest rate and future value. The field "future value" (shown in gray) must not be altered by user. Amount: Year: Interest Rate: Future Value: Calculate 2. Write a program which fill the rectangle with the selected colour when button pressed.				



MADHUBEN & BHANUBHAI PATEL
INSTITUTE OF TECHNOLOGY
The Charutar Vidya Mandal (CVM) University

DEPARTMENT OF INFORMATION TECHNOLOGY
A.Y. 2025-26, EVEN TERM
SUBJECT CODE: 202044502
SUBJECT NAME: PROGRAMMING WITH JAVA
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